

Unit 3 :: Complex Analysis

☞ **PHY0400404: Mathematical Physics** ☞

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Syllabus

Unit 3: Complex analysis

Functions of complex variables, analyticity and Cauchy-Riemann conditions, examples of analytic functions; singular functions: poles and branch points, order of singularity; integration of functions with complex variable, Cauchy's integral theorem and Cauchy's integral formula, simply and multiply connected regions; Laurent and Taylor's series expansions, residue theorem with application.

1 Functions of complex variables

1.1 Complex analytic functions

1.2 Cauchy-Riemann conditions

1.3 Examples of analytic functions

2 Singularities of complex functions

2.1 Poles

2.2 Branch points

3 Complex integration

3.1 Simply and multiply connected regions

3.2 Cauchy's integral theorem

3.3 Cauchy's integral formula

4 Complex series

4.1 Taylor's series expansion of a complex function

4.2 Laurent's series expansion of a complex function

4.2.1 Residue theorem